

National University of Computer and Emerging Sciences



Lab Manual 07 Computer Organization and Assembly Language Lab

Department of Computer Science

FAST-NU, Lahore, Pakistan

Lab Manual – String Instructions

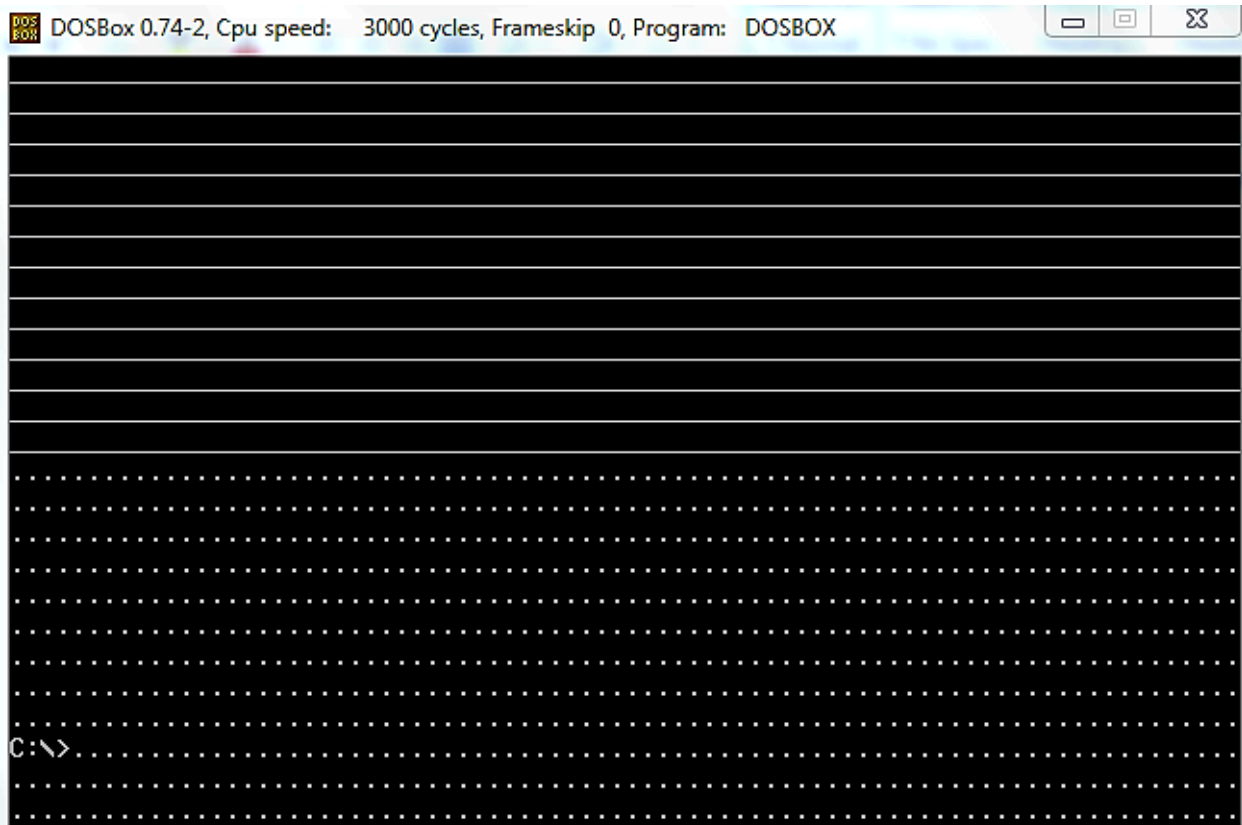
Important Instructions:

- 1- Make proper subroutines**

2- Use string instructions wherever possible in your code

Activity 1: Code to clear screen is given in example 7.1. Your task is to modify this code and print '_' (underscore) on first 13 rows of screen and '.' In rest of the rows. Required output is given below. Properly calculate the cells required with each character.

Required Output: (You are not required to explicitly print the Command Line)



Activity 2: Write a program that makes the screen constantly scroll up and down without losing any data.

Required Output:

<https://drive.google.com/file/d/129YfbKGeW02XNx9sfb-ogBJD22yHAON0/view?usp=sharing>

Practice Problems

Activity 3: Update program written in activity 1 such that it prints Introduction Screen and Activity1 Screen one by one each for 1 second in an infinite loop.

Question 1: Write a program that takes a c-string *myStr* and two characters *charToFind* and *charToReplace* and replaces all the occurrences of *charToFind* with *charToReplace* in *myStr*.

Sample output:

InputString: ddsdfhgrtsdfhjghksdd

CharToFind: d

CharToReplace: \$

ModifiedString: \$\$\$sfhgrrts\$fhghjks\$\$

Question 2: Write a program that takes a character *ch* and a CString *myStr* and removes all the occurrences of *ch* from *myStr*.

Sample Output:

myStr: cabccdefcfdcxyzcc

ch: 'c'

Modified String: abdeffd xyz

Question 3: TrimStart(char* str)

Write a function that takes a string and removes all the space in start of the string.

Sample Output:

Before TrimStart

str: " Hello How are you?"

After TrimStart

str: "Hello How are you?"

Question 4: String Compression

Write a function that compresses a string by removing consecutive occurrences of same character.

Sample Run:

String Before Compression:

Str: "ggggdddddyyyakxxuww"

String after Compression:

Str: "gdyakxuw"

Question 5: Write a function that searches a substring from a string and highlights the found substring. If the string is not found it will not highlight anything.

Sample Run:

String: "I am a student of COAL"

Substring: "student"

Printed String after Search: "I am a student of COAL"

Question 6: Write a program that finds total number of occurrences of a character from a null terminated string. For example total occurrences of 'a' in string "I am a student of coal" are 3.

Question 7: Write a program that prints tokens of a string.

Sample Run:

String: I am a student of coal

Output:

I

am

a

student

of

coal

Question 8: Write a program that flips the screen.